

AI Insight Engine — HR Configuration

Status: Specification only — not yet implemented. Epic 4 was deferred on 2026-04-15.

HR-specific configuration for the AI Insight Engine. This document is self-contained: a junior developer can read it top-to-bottom and understand the full HR AI Insight specification. Shared platform patterns live in [doc 10 \(10-ai-insight-base.md\)](#) and are referenced as "See doc 10, §X" where needed — you do not need to read doc 10 unless you want deeper platform-level context.

1. Overview

The HR configuration plugs into the base AI Insight Engine (doc 10) to provide automated analysis across seven dashboard pages:

#	Page	Route	Sections	Components
1	Workforce	/hr-dashboard/workforce	2	4
2	Attendance	/hr-dashboard/attendance	3	7 (includes OT-Sales correlation)
3	Leave	/hr-dashboard/leave	3	7
4	Performance	/hr-dashboard/performance	1	3
5	Disciplinary	/hr-dashboard/disciplinary	2	3
6	Probation	/probation	1	2
7	Payroll	/hr-dashboard/payroll	2	5 (page not yet built)

Totals: 14 sections, 31 components across 7 pages.

Excluded: Employee Directory (Workforce — searchable list, no analysis value), Probation Settings (config page, no data to analyze).

How it works (simplified):

1. User clicks "Analyze" on a section header.
2. The orchestrator fetches pre-computed data from PostgreSQL (with PII stripped and thresholds pre-flagged).
3. Phase 1: Each component is analyzed in parallel by a fast model (Haiku) using the component prompt + data.
4. Phase 2: A summary model (Sonnet) reads all component results and produces 1-6 structured insight cards (good + bad findings).
5. Results are stored in PostgreSQL (keyed per user for RBAC scoping) and displayed in the UI.

See doc 10, §2 for the full architecture diagram and §4 for the analysis lifecycle.

2. Analysis Persona

The global system prompt is prepended to every component-level LLM call. It defines who the AI is, how it should write, and what rules it must follow.

Full Global System Prompt

"You are an HR analyst reviewing the HR dashboard for a Malaysian company (Hoi-Yong). You are explaining what you see to HR management."

Prompt Rules

Rule	Detail
Language	Direct, concise, no HR jargon unless necessary
Format	Bullet points for observations, markdown tables for comparisons
Component word limit	Max 150 words
Summary detail word limit	220–320 words
Verbatim-copy rule	Every number must match a value from the data block (display rounding OK)
Scope discipline	Period-based vs snapshot — language must match the scope type
Self-verification	Cross-check numbers, arithmetic, and scope before writing
PII ban	Never reference employee names, IDs, IC numbers — columns containing PII are never whitelisted. Use department-level or aggregate language only
Threshold awareness	Fetchers pre-flag data against configurable thresholds. AI sees "flagged" vs "not flagged" — do not re-evaluate raw numbers against thresholds

3. Scope Assignments

Each section uses one of three scope types. The scope controls what language the AI uses when describing numbers and how date filtering works. See doc 10, §13 for scope type definitions.

Scope	Sections	How determined
daily	attendance_daily	Date picker on page
period	workforce_movement, attendance_monthly, attendance_flagged, leave_application, leave_analysis, performance_full, disciplinary_analysis, payroll_overview, payroll_breakdown	Year/month selector on page
snapshot	workforce_demographic, leave_balance, disciplinary_records, probation_full	Current state, no date filter

Scope follows the user's current page selection — AI analyzes what the user sees.

4. Dual-Model Strategy (HR)

HR uses the shared dual-model strategy from the base platform. See doc 10, §3 for full details.

Phase	Model	Max Tokens	Purpose	HR notes
Phase 1 — Component analysis	Haiku 4.5	2,048	Fast, parallel analysis of each component (no tools, 150-word cap)	31 components across 14 sections. Haiku keeps cost low.
Phase 2 — Summary synthesis	Sonnet 4.6	4,096	Synthesis of all component results into insight cards	Has tool access for root-cause investigation per section policy (see §7).

5. Section & Component Catalog

This section documents all 14 sections and their 31 components. For each section you will find:

- 1. A metadata table (section key, page, scope, tool policy, data sources, component count)
- 2. A component table (key, name, type, what it measures, thresholds)
- 3. The exact component prompt for each component, shown in a blockquote

5.1 Workforce — Demographic

Section Key	workforce_demographic			
Page	workforce			
Scope	snapshot			
Tool Policy	aggregate_only			
Data Sources	employee_list			
Components	3			
Key	Name	Type	What It Measures	Thresholds
wf_kpis	Workforce KPIs	kpi	Total Employees, Active Departments, Average Tenure (years)	Tenure <2yr avg = high turnover signal, >5yr = stable
wf_pie_chart	Overall Demographics	chart	Distribution by Gender, Nationality, Age, Tenure, Group	Any single category >70% = low diversity flag

Key	Name	Type	What It Measures	Thresholds
wf_dept_bar	Department Breakdown	chart	Per-department headcount with demographic split	Any dept >40% of total = concentration risk

Component Prompt — wf_kpis:

You are analyzing the "Workforce KPIs" — three cards showing Total Employees, Active Departments, and Average Tenure.

What it measures: Current workforce size, organizational breadth, and employee retention proxy.

How to read it:

- Total Employees = active headcount (excludes resigned)
- Active Departments = departments with at least 1 active employee
- Average Tenure = mean years of service across all active employees

Evaluate: Is the workforce growing or shrinking? Is tenure healthy (>3 years suggests stability)?

Component Prompt — wf_pie_chart:

You are analyzing the "Overall Demographics" pie chart showing workforce distribution by a selected category (Gender, Nationality, Age, Tenure, or Group).

What it measures: Workforce composition and diversity across demographic dimensions.

Look for: Over-concentration in any single category (>70%), unusual age distribution (aging workforce vs very young), nationality composition trends.

Component Prompt — wf_dept_bar:

You are analyzing the "Department Breakdown" stacked bar chart showing headcount per department with demographic overlay.

What it measures: How employees are distributed across departments.

Look for: Departments that are disproportionately large (>40% of total), departments with very low headcount (understaffing risk), demographic imbalances within departments.

5.2 Workforce — Movement

Section Key	workforce_movement
Page	workforce
Scope	period
Tool Policy	aggregate_only
Data Sources	employee_list (join_date, resign_date)
Components	1

Key	Name	Type	What It Measures	Thresholds
wf_joiners_leavers	Joiners & Leavers Trend	chart	Monthly new hires vs resignations over 12 months	Leavers > Joiners for 3+ months = attrition concern

Component Prompt — wf_joiners_leavers:

You are analyzing the "Joiners & Leavers" trend chart showing monthly new hires and resignations over 12 months.

What it measures: Workforce growth trajectory — whether the company is gaining or losing employees net.

How to read it:

- Joiners bar above Leavers = net growth that month
- Leavers bar above Joiners = net attrition that month

Look for: Sustained attrition (3+ months leavers > joiners), seasonal hiring patterns, resignation clusters (multiple departures in same month), overall net movement direction.

5.3 Attendance — Daily Summary

Section Key	attendance_daily
Page	attendance
Scope	daily
Tool Policy	none
Data Sources	attendance_list, public_holiday
Components	2

Key	Name	Type	What It Measures	Thresholds
att_daily_kpis	Daily KPI Cards	kpi	Present, On Leave, Absent, Worked Hours, OT Hours, Punctuality Violations, Dinner Break Violations	Attendance rate <85% = concern, OT >4.5h avg = abnormal
att_daily_table	Daily Attendance Table	table	Per-employee attendance for the selected date	Evaluate aggregate patterns, not individuals (PII)

Component Prompt — att_daily_kpis:

You are analyzing 7 daily attendance KPI cards: Present (+ OT employees), On Leave, Absence, Worked Hours, OT Hours, Punctuality Violations (Late/Early), and Dinner Break Violations.

What it measures: Single-day workforce attendance health snapshot.

How to read it:

- Present / Total = attendance rate for the day
- OT employees = staff working overtime that day
- Punctuality = late arrivals + early departures
- Dinner Break = labor compliance violations

Evaluate: Is attendance rate acceptable? Are OT levels unusual for this day? Are punctuality violations concentrated or widespread?

Component Prompt — **att_daily_table**:

You are analyzing the daily attendance table showing per-employee records for the selected date.

What it measures: Individual attendance entries including clock-in/out times, lateness, OT status.

Note: Employee names/IDs are not included. Analyze aggregate patterns only — e.g., "X employees were late by >30 minutes" rather than naming individuals.

Look for: Clusters of lateness in specific departments, unusually long/short work hours, OT concentration.

5.4 Attendance — Monthly/Yearly Summary

Section Key	attendance_monthly
Page	attendance
Scope	period
Tool Policy	aggregate_only
Data Sources	attendance_list
Components	4

Key	Name	Type	What It Measures	Thresholds
att_monthly_kpis	Monthly Summary KPIs	kpi	Present Days, Leave Days, Off Days, Absence Days, OT Days/Hours, Violations	Month avg attendance <85% = concern
att_trend_charts	Attendance & Hours Trends	chart	Two line charts: daily attendance rate over the month + worked/OT hours over the month	Declining attendance trend = flag, OT spike = flag
att_monthly_table	Monthly Summary Table	table	Per-employee monthly breakdown (days worked, leave, absent, OT, violations)	Analyze department-level patterns, not individuals

Key	Name	Type	What It Measures	Thresholds
ot_sales_correlation	OT vs Sales Trend	chart	Monthly OT hours alongside monthly sales revenue (cross-module data from Finance <code>pc_sales_daily</code>)	OT rising + sales rising = expected (busier). OT rising + sales flat/falling = labor efficiency concern

Component Prompt — `att_monthly_kpis`:

You are analyzing monthly/yearly summary KPI cards: Present Days, Leave Days, Off Days, Unpaid Days, Absence Days, OT Days (Rest/Holiday), Worked Hours, OT Hours, Total Hours, Punctuality Violations, Dinner Break Violations.

What it measures: Aggregated attendance health for the selected period.

Evaluate: Compare to prior period if available. Are absence rates rising? Is OT sustainable? Are violations trending up or down?

Component Prompt — `att_trend_charts`:

You are analyzing two trend line charts: (1) Daily attendance rate over the month, (2) Worked hours and OT hours over the month.

What they measure: Day-by-day attendance patterns and work effort within the selected period.

Look for: Mid-week dips, Friday/Monday absence patterns, OT spikes on specific days, correlation between low attendance and high OT (remaining staff covering).

Component Prompt — `att_monthly_table`:

You are analyzing the employee monthly summary table showing per-employee attendance aggregates for the period.

What it measures: Individual-level monthly totals (days worked, leave taken, absences, OT, violations).

Note: Employee names/IDs are not included. Focus on department-level patterns and statistical distributions — e.g., "the top quartile of OT hours is concentrated in 2 departments."

Component Prompt — `ot_sales_correlation`:

You are analyzing a cross-module comparison: monthly OT hours (from HR) vs monthly sales revenue (from Finance).

What it measures: Whether overtime trends correlate with business activity (sales volume).

How to read it:

- OT rising + Sales rising = business is getting busier (expected, healthy OT)
- OT rising + Sales flat or falling = possible labor efficiency problem (doing more OT without generating more revenue)

- OT falling + Sales rising = improved productivity or understaffing risk

Data source: OT hours from `attendance_list`, sales revenue from Finance's `pc_sales_daily` (monthly totals).

5.5 Attendance — Flagged Staff

Section Key	<code>attendance_flagged</code>			
Page	attendance			
Scope	period			
Tool Policy	full			
Data Sources	<code>attendance_list</code> , <code>hr_settings</code> (thresholds), <code>pattern-detection.service</code>			
Components	1			
Key	Name	Type	What It Measures	Thresholds
att_flagged_all	Flagged Employees Summary	table	Combined view of all 5 alert types: High Absence, Chronic Lateness, Early Departure, Break Compliance, Abnormal OT	Flagged/not-flagged is pre-computed by backend using configurable thresholds

Component Prompt — `att_flagged_all`:

You are analyzing the "Flagged Staff" section which combines 5 behavioral alert categories: High Absence, Chronic Lateness, Early Departure, Break Compliance Violations, and Abnormal OT.

What it measures: Employees whose attendance patterns exceed configurable thresholds (pre-flagged by the system).

The data shows: count of flagged employees per category, department distribution of flags, and trend vs prior period.

Note: Thresholds are configured by HR admin. The data is pre-flagged — you do not need to re-evaluate individual records.

Look for: Which alert category has the most flags? Are flags concentrated in specific departments? Is the total flag count growing or shrinking vs prior period? Do any employees appear in multiple categories (multi-flag risk)?

5.6 Leave — Application

Section Key	<code>leave_application</code>
Page	leave

Scope	period			
Tool Policy	aggregate_only			
Data Sources	leave_transaction			
Components	3			
Key	Name	Type	What It Measures	Thresholds
lv_app_kpis	Leave KPIs	kpi	Pending Approval count, Upcoming Leave count (Today + Future)	Pending >10 = backlog concern
lv_upcoming_table	Upcoming Leave	table	Approved leaves in the near future with dates, days, and timing	Coverage gaps (multiple same-dept leaves overlapping) = risk
lv_applications_table	Leave Applications	table	Filtered leave applications with status, type, holiday adjacency	High rejection rate (>15%) = process issue warnings

Component Prompt — lv_app_kpis:

You are analyzing 2 leave KPI cards: Pending Approval and Upcoming Leave (split into Today and Future).

What it measures: Leave pipeline health — how many applications are awaiting decision and how many approved leaves are coming.

Evaluate: Is the pending backlog reasonable? Are there many leaves upcoming in the near term (coverage risk)?

Component Prompt — lv_upcoming_table:

You are analyzing the "Upcoming Leave" table showing approved future leave entries.

What it measures: Short-term workforce availability — who is going on leave and when.

Note: Employee names are not included. Focus on department coverage — e.g., "3 employees from the same department are on leave next week."

Look for: Overlapping leaves in the same department, long-duration leaves (>5 days), clustering around holidays.

Component Prompt — lv_applications_table:

You are analyzing the leave applications table with status breakdown (Applied, Approved, Rejected, Cancelled).

What it measures: Leave application patterns and approval health.

The data includes holiday adjacency flags (leaves taken next to public holidays).

Look for: High rejection rates, leave types with unusual patterns, holiday-adjacent clustering, departments with disproportionate leave volume.

5.7 Leave — Analysis

Section Key	leave_analysis			
Page	leave			
Scope	period			
Tool Policy	full			
Data Sources	leave_transaction, leave_balance, public_holiday, hr_settings			
Components	3			
Key	Name	Type	What It Measures	Thresholds
lv_monthly_breakdown	Monthly Leave Breakdown	chart	Donut (overall distribution) + Stacked bar (monthly trend by type or status)	Sick leave >30% of total = health concern, Unpaid >10% = financial signal
lv_utilization_outliers	Leave Utilization Outliers	table	Statistical outlier detection — high usage and low usage employees per leave type	High/low thresholds from hr_settings (configurable). Pre-flagged.
lv_near_ph_patterns	Near-PH Leave Patterns	table	Employees who frequently take leave adjacent to public holidays	Flagged if adjacent count > threshold (configurable, default 3). Pre-flagged.

Component Prompt — lv_monthly_breakdown:

You are analyzing the leave monthly breakdown: a donut chart showing overall distribution and a stacked bar chart showing monthly trend (grouped by leave type or status).

What it measures: How leave is distributed across types and months.

Look for: Seasonal patterns (e.g., December spike), dominant leave types, months with unusually high leave volume, trend direction (increasing or decreasing leave usage).

Component Prompt — lv_utilization_outliers:

You are analyzing leave utilization outliers — employees flagged as statistically high or low leave users per leave type.

What it measures: Employees whose leave usage deviates significantly from the population average (using standard deviation).

The data shows: average usage %, standard deviation, high/low thresholds, and flagged employees per category.

Note: Employees are pre-flagged by the system. Names are stripped. Focus on: how many are flagged, which leave types have the most outliers, department concentration of outliers.

High usage may signal burnout or abuse. Low usage may signal presenteeism or unused entitlement risk.

Component Prompt — lv_near_ph_patterns:

You are analyzing the "Frequent Near-PH Taker" table — employees who frequently take leave adjacent to public holidays.

What it measures: Pattern of leave-taking around public holidays, which may indicate leave abuse (extending holidays).

The data shows: flagged employees with instance count, last occurrence, and associated holiday. Pre-flagged by the system against configurable threshold.

Look for: Total flagged count, department concentration, frequency of pattern (one-time vs habitual), which public holidays are most "extended."

5.8 Leave — Balance

Section Key	leave_balance
Page	leave
Scope	snapshot
Tool Policy	aggregate_only
Data Sources	leave_balance
Components	1

Key	Name	Type	What It Measures	Thresholds
lv_balance_summary	Leave Balance Summary	table	Per-employee balance by leave type: BF + Entitled + Credits - Taken - Pending = Balance	Near-zero balance = risk of unpaid leave; High unused balance = forfeiture risk

Component Prompt — lv_balance_summary:

You are analyzing the leave balance summary table showing per-employee balances across all leave types (AL, CL, HL, MC, PL, RL, ML).

What it measures: Current leave entitlement health — how much leave employees have remaining.

Note: Employee names are not included. Analyze aggregate patterns:

- How many employees have near-zero Annual Leave balance (may need unpaid leave)?
- How many have high unused balances (forfeiture risk if not taken before year-end)?
- Are any leave types being underutilized company-wide?

Formula per type: Brought Forward + Entitled + Credits - Taken - Pending = Balance

5.9 Performance — Full Page

Section Key	performance_full			
Page	performance			
Scope	period			
Tool Policy	full			
Data Sources	appraisal, appraisal_form_template			
Components	3			
Key	Name	Type	What It Measures	Thresholds
pf_kpis	Performance KPIs	kpi	Average Score, High Performers count, Low Performers count, Completion Rate (approved/total)	High Performer threshold from hr_settings (default 0.85), Low Performer threshold (default 0.6)
pf_table	Performance Table	table	Employee-level scores with previous score, current score, change direction, status	Analyze department-level distributions, not individuals
pf_charts	Performance Charts	chart	Score distribution (horizontal bar) + Average score by department (line with overall avg reference)	Department avg below overall avg = underperforming dept

Component Prompt — pf_kpis:

You are analyzing 4 performance KPI cards: Average Score, High Performers count, Low Performers count, and Completion Rate.

What it measures: Overall appraisal health for the selected year.

How to read it:

- Average Score = mean appraisal score across all evaluated employees
- High/Low Performers = employees above/below configurable thresholds (pre-flagged)
- Completion Rate = approved appraisals / total expected (includes Pending and Not Appraised counts)

Evaluate: Is the average score healthy? Is the completion rate on track? What is the ratio of high to low performers?

Component Prompt — pf_table:

You are analyzing the employee performance table showing appraisal scores, YoY change direction, and completion status.

What it measures: Individual appraisal outcomes for the selected year.

Note: Employee names are not included. Focus on:

- Distribution of score changes (how many improved vs declined vs new)
- Department-level patterns (which depts are improving/declining)
- Completion gaps (how many are "Not Appraised" or "Pending")
- Status breakdown (Approved vs Pending vs Not Appraised)

Component Prompt — pf_charts:

You are analyzing two performance charts: (1) Score Distribution — horizontal bar showing how many employees fall in each score range, (2) Average Score by Department — line chart with overall average as reference.

What they measure: Score spread across the workforce and department-level performance comparison.

Look for: Is the distribution bell-shaped (healthy) or skewed? Are most employees clustered in a narrow range (rating inflation)? Which departments are above/below the overall average? Is there significant spread between departments?

5.10 Disciplinary — Records/KPIs

Section Key	disciplinary_records			
Page	disciplinary			
Scope	snapshot			
Tool Policy	aggregate_only			
Data Sources	hr_warnings, hr_offense_categories			
Components	2			
Key	Name	Type	What It Measures	Thresholds
dc_kpis	Disciplinary KPIs	kpi	Pending Approval, Under Coaching, At Risk (with 2nd/3rd warning sub-counts)	At Risk >0 = immediate attention, Pending >5 = process backlog

Key	Name	Type	What It Measures	Thresholds
dc_warning_list	Active Warnings	table	Current warning records for active employees with type, offence, status	Analyze by department and offence type, not individuals

Component Prompt — **dc_kpis**:

You are analyzing 3 disciplinary KPI cards: Pending Approval (warnings awaiting sign-off), Under Coaching (active coaching cases), and At Risk (employees with 2nd or 3rd stage warnings).

What it measures: Current disciplinary pipeline health.

Evaluate: Is the pending queue growing? How many employees are under active coaching? How many are at-risk (2nd/3rd warning — potential termination path)? What is the ratio of coaching to at-risk?

Component Prompt — **dc_warning_list**:

You are analyzing the active warnings table showing current warning records.

What it measures: Active disciplinary cases — what types of offences are occurring and their current status.

Note: Employee names are not included. Focus on:

- Most common offence types
- Department concentration of warnings
- Warning stage distribution (1st vs 2nd vs 3rd)
- Action types (Written Warning, Coaching, etc.)

5.11 Disciplinary — Analysis

Section Key	disciplinary_analysis
Page	disciplinary
Scope	period
Tool Policy	full
Data Sources	hr_warnings
Components	1

Key	Name	Type	What It Measures	Thresholds
dc_charts	Disciplinary Trends	chart	Donut (overall by category) + Stacked bar (monthly trend by action/offence/department)	Rising monthly trend = concern, seasonal spikes = investigate

Component Prompt — **dc_charts**:

You are analyzing the disciplinary analysis charts: a donut showing overall warning distribution (by Action Type, Offence Type, or Department) and a stacked bar showing monthly trend.

What it measures: Warning volume patterns over the selected year.

Look for: Is the monthly volume increasing or decreasing? Which category dominates? Are there seasonal patterns (e.g., post-holiday spikes)? Is one department generating disproportionate warnings? Which offence types are most common?

5.12 Probation — Full Page

Section Key	probation_full
Page	probation
Scope	snapshot
Tool Policy	aggregate_only
Data Sources	employee_list, probation_settings
Components	2

Key	Name	Type	What It Measures	Thresholds
pb_kpis	Probation KPIs	kpi	Under Probation (with Just Join / Mid Way sub-counts), Overdue (requires immediate HR action)	Overdue >0 = critical, Overdue/Total >20% = systemic process failure
pb_records_table	Probation Records	table	Employee probation status: period, join date, end date, remaining days, status, action	Remaining days <14 = urgent, negative remaining = overdue

Component Prompt — pb_kpis:

You are analyzing 2 probation KPI cards: Under Probation (split into Just Joined and Mid Way) and Overdue (probation period expired without confirmation).

What it measures: Probation pipeline health — how many new employees are being evaluated and how many have fallen through the cracks.

Evaluate: Is the overdue count zero (ideal) or growing? What percentage of probationers are overdue (process health)? Is the Just Joined vs Mid Way ratio reasonable for the company's hiring pace?

Component Prompt — pb_records_table:

You are analyzing the probation records table showing all employees currently in probation.

What it measures: Individual probation status — remaining days, current stage, and HR action taken.

Note: Employee names are not included. Focus on:

- How many are overdue (negative remaining days) — color-coded red
- How many are approaching deadline (<14 days) — color-coded orange
- Department distribution of probationers
- Action taken vs not (how many have been confirmed, discontinued, or still pending)

5.13 Payroll — Overview (Page Not Yet Built)

Section Key	payroll_overview
Page	payroll
Scope	period
Tool Policy	none (page not built, tables TBD)
Data Sources	TBD (payroll tables not yet defined)
Components	3

Key	Name	Type	What It Measures	Thresholds
pr_total_kpis	Payroll Total KPIs	kpi	Total payroll cost, headcount, avg cost per employee	YoY growth >10% = concern
pr_dept_waterfall	Department Payroll Waterfall	chart	Payroll cost breakdown by department	Any dept >40% of total = concentration
pr_yoy_trend	Payroll YoY Trend	chart	Monthly payroll cost over 12 months with YoY comparison	Rising faster than headcount growth = cost pressure

PII Rule: Department averages only. AI never sees individual salaries.

Component Prompt — payroll_overview:

You are analyzing payroll KPI cards: Total Payroll Cost, Headcount, and Average Cost Per Employee.

What it measures: Overall payroll spend and per-head cost for the selected period.

Evaluate: Is total payroll growing? Is per-head cost rising (salary inflation) or stable? How does headcount change relate to total cost change?

Component Prompt — pr_dept_waterfall:

You are analyzing a waterfall chart showing payroll cost broken down by department.

What it measures: How payroll spend is distributed across departments.

Look for: Which department has the highest payroll cost? Is any department disproportionately expensive relative to its headcount? Any unexpected changes vs prior period?

Component Prompt — **pr_yoy_trend**:

You are analyzing a 12-month payroll trend with year-over-year comparison.

What it measures: Monthly payroll cost trajectory and how it compares to the same months last year.

Look for: Is payroll trending up or down? Are there seasonal spikes (bonuses, year-end)? Is YoY growth consistent or accelerating?

5.14 Payroll — Breakdown (Page Not Yet Built)

Section Key	payroll_breakdown			
Page	payroll			
Scope	period			
Tool Policy	none (page not built, tables TBD)			
Data Sources	TBD (payroll tables not yet defined)			
Components	2			
Key	Name	Type	What It Measures	Thresholds
pr_component_breakdown	Payroll Component Breakdown	chart	Split of payroll into base salary, OT pay, allowances, claims	OT pay >20% of total in any dept = flag
pr_cost_drivers	Payroll Cost Drivers	table	Top cost-increasing factors vs prior period	Any single factor >50% of total increase = concentrated

PII Rule: Department averages only. AI never sees individual salaries.

Component Prompt — **pr_component_breakdown**:

You are analyzing payroll component breakdown: base salary, OT pay, allowances, and claims as percentages of total payroll.

What it measures: Where payroll money goes — fixed vs variable components.

Look for: Is OT pay growing as a share of total? Are allowances or claims spiking? Which component changed the most vs prior period?

Component Prompt — **pr_cost_drivers**:

You are analyzing the payroll cost drivers table showing what caused payroll to increase or decrease vs prior period.

What it measures: Root causes of payroll changes.

Look for: Is the increase driven by headcount growth, salary adjustments, or OT? Which department contributed most to the change?

6. Deterministic Summary Questions

Each section has fixed questions the AI must answer during summary synthesis. This makes the summary output predictable — same data always answers the same questions. See doc 10, §16 for the shared pattern.

Section	Summary Questions
workforce_demographic	1. Is headcount growing or shrinking vs last month? 2. Is average tenure above 3 years? 3. Any department with less than 3 staff?
workforce_movement	1. How many joiners vs leavers this month? 2. Is there a department losing more people than others? 3. What's the turnover rate?
attendance_daily	1. What's today's attendance rate? Is it below 85%? 2. How many staff are late today? 3. How many are on OT?
attendance_monthly	1. Is monthly avg attendance above 85%? 2. Is OT hours trending up or down? 3. Does OT trend match sales trend (cross-module check)? 4. Any department with attendance below 80%?
attendance_flagged	1. How many staff are flagged for chronic lateness? 2. How many flagged for post-public-holiday leave? 3. How many flagged for abnormal OT (>4.5h for 10+ days)?
leave_application	1. How many leave applications this month vs last month? 2. What's the approval rate? 3. Any spike in a specific leave type?
leave_analysis	1. Which leave type is most used? 2. Any staff taking leave right before/after public holidays repeatedly? 3. Any department using more leave than average?
leave_balance	1. How many staff have used less than 30% of annual leave? 2. How many are at risk of forfeiting leave (>80% remaining near year end)?
performance_full	1. What's the average appraisal score? Is it above 0.7? 2. How many high performers (>0.85) vs low performers (<0.6)? 3. Any department with avg score below overall average?
disciplinary_records	1. How many active warnings currently? 2. Which department has the most warnings? 3. What's the most common offence type?
disciplinary_analysis	1. Are warnings increasing or decreasing vs last month? 2. Any staff with 2+ warnings in the same period? 3. Any pattern in timing (e.g., more

Section	Summary Questions
	warnings on certain days)?
probation_full	1. How many staff currently on probation? 2. Any probation reviews overdue? 3. Any reviews due in the next 2 weeks?
payroll_overview	1. What's total payroll cost this month? Up or down vs last month? 2. Which department has the highest payroll cost? 3. Is YoY payroll growth above 10%?
payroll_breakdown	1. What % of payroll is base salary vs OT pay vs allowances? 2. Which component grew the most vs last month? 3. Any department where OT pay exceeds 20% of total payroll?

7. Tool Policy

The tool policy controls whether the LLM can make database queries during analysis. See doc 10, §7 for the shared tool-use system.

Design

HR uses a single tool `query_hr_table` — all data is in local PostgreSQL (no remote RDS like Finance). The tool enforces column whitelisting and a `ROW_LIMIT = 100` per query. The AI can query any date range or filter, but results are capped at 100 rows.

Only Sonnet (summary synthesis) gets tool access. Haiku (component narration) never has tools.

Three Tiers

Tier	What the LLM Can Do	When Used
none	No tool access. LLM receives only the pre-fetched data block.	Single-day attendance (all data pre-fetched), payroll (page not built).
aggregate_only	Can query <code>HR_AGGREGATE_TABLES</code> only via <code>query_hr_table</code> .	Overview/snapshot sections — data is already pre-aggregated, tools for light drill-down.
full	Can query all whitelisted tables via <code>query_hr_table</code> .	Analysis/flagged sections — used for root-cause investigation.

Per-Section Assignments

Section Key	Policy	Rationale
workforce_demographic	aggregate_only	Snapshot KPIs — drill into dept-level counts
workforce_movement	aggregate_only	Period trend — joiners/leavers by month
attendance_daily	none	Single day, all data pre-fetched

Section Key	Policy	Rationale
attendance_monthly	aggregate_only	Period aggregates — attendance trends, OT patterns
attendance_flagged	full	Drill into specific flag categories, dept breakdown
leave_application	aggregate_only	Period stats — approval rates, type distribution
leave_analysis	full	Outlier investigation needs full table access
leave_balance	aggregate_only	Snapshot — balance distributions by type
performance_full	full	Score distributions, dept comparisons, YoY changes
disciplinary_records	aggregate_only	Snapshot — active warning counts by dept/type
disciplinary_analysis	full	Trend investigation, multi-warning patterns
probation_full	aggregate_only	Snapshot — overdue/upcoming counts
payroll_overview	none	Page not built, tables TBD
payroll_breakdown	none	Page not built, tables TBD

Pattern: Snapshot/overview sections use **aggregate_only**. Analysis/flagged sections use **full**. Payroll and daily attendance use **none**.

8. Data Source Tables

All HR data lives in Prisma-managed PostgreSQL (no remote RDS like Finance).

Table	Modules	Key Columns
employee_list	Workforce, Probation	employee_code, name, department, join_date, resign_date, is_active
attendance_list	Attendance	date, employee_code, clock_in, clock_out, lateness, status, ot_hours
leave_transaction	Leave	employee_code, leave_type, start_date, end_date, days, apply_status, is_adjacent_holiday
leave_balance	Leave	employee_code, leave_type, bf, entitled, credits, taken, pending, balance
public_holiday	Attendance, Leave	date, name
appraisal	Performance	employee_code, year, score, status, department
appraisal_form_template	Performance	criteria, weights
hr_warnings	Disciplinary	employee_code, warning_stage, offence_category, action_type, status,

Table	Modules	Key Columns
		department
hr_offense_categories	Disciplinary	category, description
probation_settings	Probation	department, months
hr_settings	All (thresholds)	category, key, value

9. Configurable Thresholds (from hr_settings)

Unlike Finance (hardcoded thresholds), HR thresholds are configurable via the Settings UI and stored in hr_settings.

Design: Fetchers pre-flag data using thresholds. The AI sees flagged/not-flagged results. Thresholds are NOT injected into prompts — the fetcher does the evaluation.

Category	Used By Section	Default Values
alert_chronic_lateness	attendance_flagged	chronic_lateness_threshold = 3
alert_high_absence	attendance_flagged	high_absence_threshold = 3
alert_early_departure	attendance_flagged	early_departure_threshold = 3
alert_break_compliance	attendance_flagged	threshold = 3
alert_labor_compliance	attendance_flagged	daily_work_hour_limit = 12h, violation_threshold = 3, dinner_break_window_start = 17:00, dinner_break_window_end = 19:00, dinner_break_min_duration = 30m, ot_dinner_break_cutoff = 21:00
alert_abnormal_ot	attendance_flagged	abnormal_ot_hour_threshold = 4.5h, abnormal_ot_flag_threshold = 10 days
alert_leave_pattern	leave_analysis	leave_abuse_threshold = 3, adjacent_holiday_days = 2
alert_leave_utilization	leave_analysis	leave_utilization_low_threshold = 0.3, leave_utilization_high_threshold = 0.9
analytics_appraisal	performance_full	top_performer_threshold = 0.85, low_performer_threshold = 0.6
alert_at_risk	disciplinary_records	atrisk_weight_warning = 0.5, atrisk_weight_appraisal = 0.3, atrisk_weight_combined = 0.2, atrisk_threshold = 0.3, poor_appraisal_threshold = 0.625

10. Column Whitelisting (Data Protection)

HR uses the same column whitelisting strategy as Finance. See doc 10, §15 for the shared pattern. PII-containing columns are never whitelisted — the LLM cannot access them.

HR Column Whitelist

Each HR table declares allowed columns. Only these columns appear in fetcher output and tool queries.

Table	Whitelisted Columns
employee_list	department_code, department_description, job_title, employee_type_code, employee_group_title, gender, nationality_description, is_foreigner, is_active, join_date, confirm_date, resign_date, branch_code, branch_description, wages_type, is_disable
attendance_list	date, work_type, status, shift, start_work, end_work, hour_worked, ot, lateness, early_out
attendance_upload	uploaded_at, row_count, employee_count, date_range_start, date_range_end, status
leave_transaction	leave_date, leave_type_code, leave_type_description, apply_date, apply_status, apply_status_description, day_no, hour_no, days, is_hourly, is_credit, is_adjustment, is_adjacent_holiday, adjacent_holiday_name, days_from_holiday
leave_balance	leave_type, bf, entitled, credits, expiring_credits, expired_credits, taken, available, pending, balance_year, balance_month
public_holiday	date, holiday_name, location, is_active
appraisal	appraisal_year, department, job_title, appraised_date, rating_1–rating_20, final_score, status, hr_validated, director_validated, employee_acknowledged
appraisal_form_template	form_name, in_use, criteria_1_name–criteria_20_name, criteria_1_weight–criteria_20_weight
hr_warnings	department_code, department_description, incident_date, stage, stage_name, offense_category, status, di_required, final_decision, version, created_at
hr_offense_categories	name, examples, is_default, is_active

Aggregate vs Full Table Access

Tables allowed under **aggregate_only** policy (reference + high-level data):

- employee_list

- `attendance_list`
- `leave_balance`
- `leave_transaction`
- `public_holiday`
- `hr_offense_categories`

Tables only available under `full` policy (granular drill-down):

- `appraisal`
- `appraisal_form_template`
- `hr_warnings`
- `attendance_upload`

PII Exclusions — Never Whitelisted

- **Names:** `first_name`, `last_name`, `full_name`, `employee_name`
- **IDs:** `employee_code`, `employee_id`, `ic_number`, `passport_number`
- **Contact:** `email`, `phone_number`, `mobile_number`, `address` fields
- **Salary:** `salary`, `basic_salary`, `gross_salary`, `bank_account`, `account_number`

How It Differs From Finance

Finance whitelists customer/company names because they're needed for actionable insights ("Customer ABC owes RM 500K"). HR does **not** whitelist employee names because department-level analysis suffices ("3 employees in Warehouse are chronically late").

Additional HR Protection Layer: Pre-Flagging

Backend fetchers pre-compute flagged/not-flagged status against `hr_settings` thresholds before sending to AI. This serves two purposes:

1. **Frontend display** — flagged badges shown in the dashboard UI
2. **AI input** — AI reads flags, not raw threshold values

Runtime Enforcement

`validateColumns()` rejects any tool call requesting columns not in the whitelist. Enforced at query execution time, not prompt-level.

Prompt Guidance

Component prompts state: "Employee names/IDs are not included. Focus on aggregate/department-level patterns."

11. RBAC Scoping (Role-Based Access Control)

Role	Can Trigger Analysis	Can View Insights	Data Scope
Superadmin / HR / Director	Yes	Yes	All employees
Finance	Yes	Yes	Own department only
Manager	Yes	Yes	Own dept + direct reports
Sale / Operation	No	No (403)	Denied — no insight panel shown

Implementation (planned):

- Scope filter from `data-scoping-service.ts` injected into every fetcher query
- Different users see different data — per-user caching required (storage keyed by `user_id`)
- Client-side: `canViewInsights = userRole !== 'sale' && userRole !== 'operation'`
- Server-side: role validation on generate/latest endpoints. Unauthorized roles get 403.

See doc 10, §18 for the shared RBAC pattern.

12. Summary System Prompt

The summary system prompt is used in Phase 2 when Sonnet synthesizes all component results into insight cards. Full text:

You are an HR analyst synthesizing dashboard data for HR management. Produce insights using the `===INSIGHT===` format.

Rules:

- Max 3 good + 3 bad insights, ranked by business impact
- Every number must trace to the raw data blocks below
- Use department-level language, never reference individual employees
- For "bad" insights: name specific departments, time periods, or categories with numbers
- For "good" insights: name the drivers of positive trends

Detail structure (mandatory sections):

1. **Current Status** — headline number + business meaning
2. **Key Observations** — 2-4 bullets with non-obvious patterns
3. **Supporting Evidence** — markdown table (3+ rows) of top contributors, or 3-5 bullets of specific numbers
4. **Implication** — 1-2 bullets on business consequence

See doc 10, §12 for the shared `===INSIGHT===` output format and sentiment mapping.

13. Storage DDL

HR-specific database tables. Same structure as Finance with `user_id` dimension for RBAC-scoped caching.

```
CREATE TABLE hr_ai_insight_section (
  id SERIAL PRIMARY KEY,
  module TEXT NOT NULL,
  section_key TEXT NOT NULL,
  summary_json JSONB NOT NULL,
  analysis_time_s REAL,
  token_count INT,
  cost_usd REAL,
  period_start TEXT,
  period_end TEXT,
  user_id TEXT NOT NULL,
  generated_by TEXT NOT NULL,
  generated_at TIMESTAMPTZ DEFAULT NOW(),
  UNIQUE(module, section_key, user_id)
);

CREATE TABLE hr_ai_insight_component (
  id SERIAL PRIMARY KEY,
  section_id INT REFERENCES hr_ai_insight_section(id) ON DELETE CASCADE,
  component_key TEXT NOT NULL,
  component_type TEXT NOT NULL,
  analysis_md TEXT,
  token_count INT,
  generated_at TIMESTAMPTZ DEFAULT NOW(),
  UNIQUE(section_id, component_key)
);
```

Concurrency lock uses the same global lock pattern as Finance. See doc 10, §5 for the shared lock schema and stale expiry logic.

14. API Routes

```
GET /api/v1/hr/ai-insight/:module/insights/latest    -- cached or {
cached: false }
POST /api/v1/hr/ai-insight/:module/insights/generate -- SSE stream
```

SSE events: `progress`, `complete`, `error`, `cancelled` (same as Finance — see doc 10, §9 for the shared SSE pattern).

15. File Structure

Planned file structure for the HR AI Insight module:

```

code/backend/src/modules/hr/ai-insight/
  types.ts           -- HR-specific types
  client.ts          -- Anthropic client config (Haiku + Sonnet)
  prompts.ts         -- Registry + 31 component prompts + summary prompt
  data-fetcher.ts    -- 31 fetcher functions
  orchestrator.ts    -- Phase 1 (parallel components) + Phase 2
(summary)
  numeric-guard.ts   -- Whitelist validation (copy from Finance + add
hours unit)
  storage.ts         -- PostgreSQL read/upsert with user_id
  lock.ts           -- Per-user-per-module lock
  column-whitelist.ts -- HR_LOCAL_WHITELIST + validateColumns()
  settings-loader.ts -- Batch-fetch hr_settings for threshold pre-
flagging
  component-info.ts  -- Static "about" content per component
  response-adapter.ts -- SummaryJson → E2E-expected format

```

16. Implementation Sequence

Build order for a developer implementing the HR AI Insight module:

Step	What	Depends On
1	<code>types.ts</code> + <code>client.ts</code> — copy from Finance, adapt types	Nothing
2	<code>prompts.ts</code> — registry (14 sections, 31 components) + all component prompts + summary prompt	<code>types.ts</code>
3	<code>settings-loader.ts</code> — batch query <code>hr_settings</code> by categories	Database access
4	<code>column-whitelist.ts</code> — define <code>HR_LOCAL_WHITELIST</code> per table + <code>validateColumns()</code>	<code>types.ts</code>
5	<code>data-fetcher.ts</code> — 31 fetchers (start with Workforce, one module at a time)	<code>settings-loader.ts</code> , <code>column-whitelist.ts</code>
6	<code>numeric-guard.ts</code> — copy from Finance, add <code>hours</code> unit	Nothing
7	<code>orchestrator.ts</code> — copy from Finance, remove RDS tool-use logic, add scope/PII threading	All above
8	<code>storage.ts</code> + <code>lock.ts</code> — adapt Finance pattern with <code>user_id</code> dimension	Storage tables (DDL from §13)
9	<code>response-adapter.ts</code> — map good/bad sentiment to output format	<code>types.ts</code>
10	API routes — controller + SSE streaming	<code>orchestrator.ts</code> , <code>lock.ts</code> , <code>storage.ts</code>

Step	What	Depends On
11	<code>component-info.ts</code> — static "about" content	Nothing
12	Frontend — InsightSectionHeader + AilInsightPanel per section	API routes, client.ts

17. Verification Plan

Per-Section Smoke Test

For each of the 14 sections:

1. Click "Analyze" on the section header.
2. Verify SSE stream shows progress events for each component.
3. Verify all components complete (check count matches the catalog in §5).
4. Verify summary produces 1-6 insight cards (good + bad).
5. Verify cached results load on page refresh (GET latest endpoint).
6. Verify component detail dialog shows markdown analysis.

Fetcher / PII Verification

1. Unit test each fetcher — returns `{ prompt, allowed[] }`, PII stripped, thresholds applied.
2. Column whitelist test — no PII columns in any fetcher output or tool query result.

Tool Policy Verification

1. For `none` sections: verify no tool calls are made.
2. For `aggregate_only` sections: verify tool calls only access `HR_AGGREGATE_TABLES`.
3. For `full` sections: verify tool calls can access all whitelisted tables.
4. Verify `validateColumns()` rejects queries with non-whitelisted columns.

Threshold Verification

For each threshold in §9: provide data that triggers the flag and verify the fetcher pre-flags it correctly. The AI should see "flagged" — not re-evaluate raw numbers.

RBAC Verification

1. Log in as Superadmin/HR/Director: verify "Analyze" button is visible, all data returned.
2. Log in as Finance/Manager: verify scoped data (own department only).
3. Log in as Sale/Operation: verify 403 response, no insight panel shown.
4. Verify different users get different cached results (per-user storage keyed by `user_id`).

Numeric Guard Verification

1. Mock AI response with hallucinated numbers — verify numeric guard catches them.

Integration / E2E

1. Integration test orchestrator — mock AI responses, verify SSE events + storage writes.

2. E2E test — run against `ai-insight-parity.spec.ts` (response schema matches).
3. Playwright browser verification — click Analyze, verify UI shows insights correctly.